Agricultural and Forest Meteorology 168 (2013) 15–25

Contents lists available at SciVerse ScienceDirect

Agricultural and Forest Meteorology

journal homepage: www.elsevier.com/locate/agrformet

Large-scale weather types, forest fire danger, and wildfire occurrence in the Alps

Clemens Wastl[a,*], Christian Schunk[a], Marvin Lüpke[a], Giampaolo Cocca[b], Marco Conedera[c], Eva Valese[d], Annette Menzel[a]

[a] Chair of Ecoclimatology, Technische Universität München, Hans-Carl-von-Carlowitz-Platz 2, D-85354 Freising, Germany
[b] Ente Regionale per i Servizi all'Agricoltura e alle Foreste, Via Taramelli 12, I-20124 Milano, Italy
[c] WSL, Swiss Federal Institute for Forest, Snow and Landscape Research, Insubric Ecosystems Research Group, Via Belsoggiorno 22, CH-6500 Bellinzona, Switzerland
[d] Dipatrimento Territorio e Sistemi Agro Forestali, Università degli Studi di Padova, Viale dell'Università 16, I-35020 Legnaro, Padova, Italy

ARTICLE INFO

Article history:
Received 13 July 2012
Received in revised form 21 August 2012
Accepted 23 August 2012

Keywords:
Alps
Synoptic weather types
Forest fire danger
Fire occurrence
Canadian Forest Fire Danger Rating System
Foehn

ABSTRACT

In the Alps forest fires have burnt around 14,500 ha per year in the past decade. In this paper we studied large-scale (synoptic) weather patterns and the corresponding occurrence of forest fires in this complex topography. The database for our analysis comprised three main parts: a daily classification of weather types in the period 1951–2010, daily calculated forest fire danger indices at five selected stations in the Alps (1951–2010) and ten years of observed forest fires (2001–2010). Firstly we analyzed the frequency of the 11 different weather types and show that the Alps are a region where cyclonic flows in general, and westerly cyclonic in particular, are the dominating large-scale weather pattern due to their location in the westerlies of the global circulation system. Comparing the weather types with three calculated sub-indices of the Canadian Forest Fire Danger Rating System (FFMC, DC, DMC) at five selected sites (representative of the different climate regions in the Alps) revealed a strong dependence of meteorological forest fire danger on flow direction and cyclonality. Cyclonic weather types were characterized by a high relative humidity and in consequence a low fire danger, while the calculated fire danger in anticyclonic weather situations was significantly higher. Furthermore, strong regional differences occurred in dependence on the flow direction. Northerly winds resulted in low fire danger north of the Alps, due to orographic enhanced precipitation, and high forest fire danger south of the Alps, because of dry katabatic foehn winds. In general, the stations in the Northern Alps showed significantly lower fire index values than the stations in the south and additionally a stronger seasonal variation with considerably higher index values in summer. Regional differences were highest for the FFMC, followed by the DMC and the DC, and could be attributed to the time lag of different forest soil layers. DMC and DC relate to a rather thick soil layer which reacts very slowly and since weather types in the Alps usually change every 7th day, drying of this deep layer is too slow to reveal significant differences between the regions. The Alpine forest fire database was analyzed on a national basis to identify correlations between observed fires and large-scale weather types. 95% of the observed fires in the EU-defined Alpine Space in the past decade occurred in the two southern countries Italy and France. This was likely due to both favorable climatic conditions and better database quality in these two countries. Unfortunately, the datasets of some regions north of the Alps (e.g. in Switzerland, Germany, Austria) were very patchy. A strong human influence on the Alpine fire regime resulted in a generally low correlation between weather types and observed forest fires. Surprisingly, many forest fires occurred in conjunction with cyclonic weather types. This could be explained by the start date of the fire which was mostly at the end of a drought period when the large-scale synoptical conditions had already turned to cyclonic. Nevertheless, most and biggest fires occurred during high pressure systems and other anticyclonic situations when the fuels were completely dry.
© 2012 Elsevier B.V. All rights reserved.

1. Introduction

Wildfires in Europe are usually associated with southern countries such as Greece, Italy, Spain or Portugal where every year millions of hectares of forests are destroyed (Giglio et al., 2006), but they can also affect the Alpine region of Central Europe. Naturally, frequency and especially intensity and size of forest fires in the Alps

* Corresponding author. Tel.: +49 08161 714743; fax: +49 08161 714753.
E-mail address: wastl@wzw.tum.de (C. Wastl).

0168-1923/$ – see front matter © 2012 Elsevier B.V. All rights reserved.
http://dx.doi.org/10.1016/j.agrformet.2012.08.011

are relatively low compared to the fire prone countries surrounding the Mediterranean Sea, but due to the high population density and sensitive ecosystems wildfires in the Alps can cause a lot of damage. The Alpine forest fire database (Valese et al., 2011) revealed that in the ten years between 2001 and 2010 on average about 2400 fires occurred per year, burning around 14,500 ha. Most fires (around 90%) were restricted to the southern slopes of the Alps from France in the West to Italy and Slovenia in the East, but also regularly including the dry inner Alpine valleys (Valais in Switzerland, Inn valley in Austria, etc.; e.g. Zumbrunnen et al., 2009).

The Alps are characterized by a very high climatic variation on a comparatively small horizontal scale. This is partly caused by their location between the Atlantic climate zone in the West, the continental climate zone in the East and the Mediterranean climate zone in the South, but also their complex topography with altitudes ranging up to 4800 m asl plays a decisive role in this context. South of the Alps some areas are characterized by warm and dry winters with frequent occurrences of katabatic foehn winds and wet summers (insubric regions), whereas other areas display wet winters and very hot and dry summers with a generally high forest fire danger. North of the Alps, precipitation occurs regularly across the whole year, temperatures in summer are moderate and are cold in winter. The inner Alpine valleys are characterized by low precipitation due to rain shadowing effects and frequently occurring katabatic foehn winds which can result in a very high forest fire danger.

Although most forest fires (more than 80%) in the Alps are induced by humans (silvicultural practices, negligence or by premeditated actions) meteorology plays an important fire-predisposing role both in terms of vegetation and fuel produced and as fire-prone meteorological conditions (high temperatures, low humidity, droughts). In addition, in regions with a very steep topography (e.g. Grisons, Ticino, Aosta Valley) and at higher elevations (near the timberline) meteorological conditions inducing lightning may be responsible for a considerable number of forest fire ignitions during the summer months (around 30% in average, up to 50% in very dry and hot summers like 2003, Conedera et al., 2006).

Forest fires are a natural disturbance which can cause extensive damage and even pose a threat to humans (Frelich, 2002; Bowman et al., 2009). Nevertheless they are not as prominently reported in the Central European media such as floods, storms or avalanches. Although its wildfire regime is quite moderate compared to other regions such as Russia, Canada, the US or Australia; forest fire research has quite a long history in Central Europe. In Bavaria (southern Germany) a forest fire database was set up and analyzed in the 1950s, resulting in a regional forest fire danger index (Baumgartner et al., 1967). South of the Alps several studies were carried out in the middle of the 20th century to assess forest fire danger (e.g. France, Gatheron and Lavoine, 1950). Most recent Alpine wildfire studies have dealt with anthropogenic influences on fire regimes (Krebs et al., 2010; Pezzatti et al., in press) or impacts of climate change on forest fire danger (Reinhard et al., 2005; Zumbrunnen et al., 2009; Wastl et al., 2012). Furthermore, fire control and fire management practices have also been a subject of research within the Alpine forest fire community (e.g. Conedera et al., 2009). Despite the prominent role of climate, however, very little research has been performed so far on how large-scale weather patterns influence the occurrence of forest fires in the Alps. Such studies only exist for more forest fire prone countries such as Canada or those in the Mediterranean region. Skinner et al. (2002), for example, combined 500 hPa synoptic patterns with the Canadian Large Fire Database and found that a large area burnt was usually associated with mid-tropospheric ridging and a low zonal index. Similar studies for the Iberian Peninsula also showed a strong correlation between certain atmospheric circulations at the synoptic scale (blocking high, southerly winds) and the occurrence of very large forest fires (Hoinka et al., 2009; Rasilla et al., 2010).

The purpose of the present paper is to establish climatological relationships between large-scale synoptic weather types over Central Europe, derived from the 'Großwetterlagenkatalog' of the German Meteorological Service, and the meteorological forest fire danger based on several sub-indices from the Canadian Forest Fire Danger Rating System (CFFDRS, Van Wagner, 1987) as well as observed forest fires. Due to the highly variable climatological conditions in the Alps the statistical analysis was done at a regional scale based on five climatic zones. A better knowledge about weather/climate influences on fire occurrence would greatly help to improve the understanding of possible impacts on the Alpine fire regime of changing flow patterns associated with global warming.

The content of our paper is as follows: after a description of the region of interest and of the meteorological and forest fire databases we briefly describe the climatology of different large-scale weather types in Central Europe. We then analyze the correlation between synoptic conditions and different forest fire danger indices as well as observed forest fires in the Alps. The paper concludes with a summary of the main findings and an outlook on forthcoming research.

2. Materials and methods

2.1. Study area and meteorological station data

The area under investigation is displayed in Fig. 1 and comprises the whole Alpine area (defined by the European Union, http://www.alpine-space.eu/) stretching over France, Switzerland, Germany, Italy, Austria and Slovenia.

Due to very diverse climate conditions and complex topography, a great number of different vegetation types and ecosystems are present in the Alpine region, ranging from Mediterranean forests in the southern parts to boreal-like conifer stands at higher altitudes in the northern parts. On a much smaller scale, e.g. across an inner Alpine valley, vegetation can vary from Scots Pine stands and grassland on the southern exposed slopes to Norway Spruce and broadleaf stands on the shady and mesic northern exposed sides.

In order to account for the different climatic conditions within the Alps we decided to refer to the regionalization of the HISTALP project of the Austrian weather service (www.zamg.ac.at/histalp), where four climate regions in the Alps (Northwest, Northeast, Southwest, Southeast) plus one additional class covering the high altitude stations in the inner Alpine region (summits) have been defined on the base of a principal components analysis of the annual records of all stations (Auer et al., 2007).

For each defined climate region we selected a representative station with a long homogeneous time series (at least the past 60 years) of all standard meteorological parameters necessary for the fire danger index calculation. Accordingly, the selected stations are Zurich (Z, region northwest, see Fig. 1) and Locarno (Lo, region southwest) in Switzerland, Linz (L, region northeast) and Innsbruck (I, inner Alpine region) in Austria and Ljubljana in Slovenia (Lj, region southeast). The elevation of these stations ranges between 260 m (Linz) and 577 m asl (Innsbruck).

The data comprise daily values (12 CET = UTC + 1 h) of air temperature, precipitation, wind speed and relative humidity for the time period 1951–2010 and are provided by MeteoSwiss (Switzerland, 2 stations), ZAMG (Zentralanstalt für Meteorologie und Geodynamik, Austria, 2 stations) and the Environmental Agency of the Republic of Slovenia (Slovenia, 1 station).

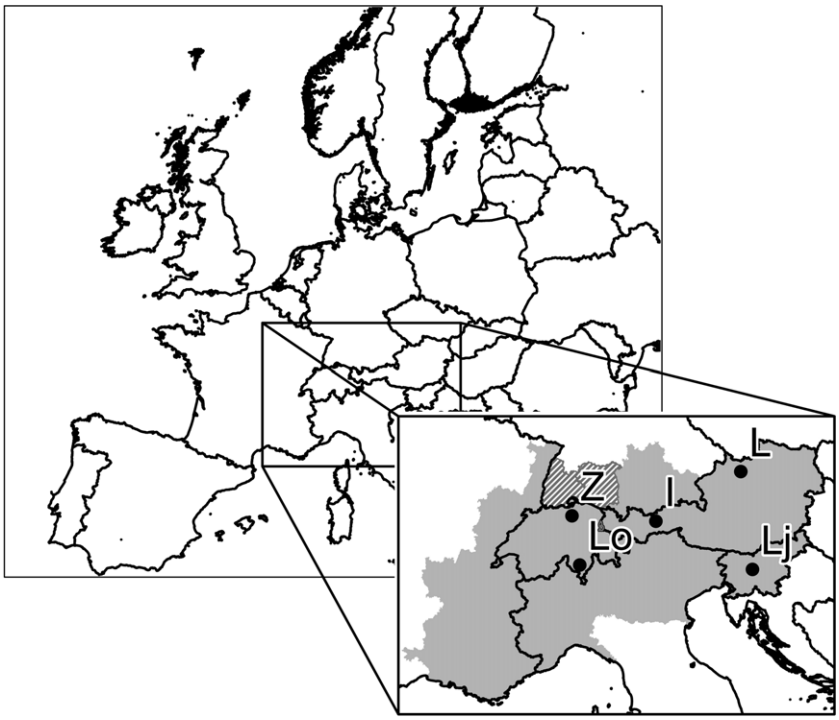


Fig. 1. Location of the five climate stations selected for the investigation. The station names, abbreviations, altitudes and the according HISTALP regions from West to East are: Zürich (Z, 556 m, region NW), Locarno (Lo, 383 m, region SW), Innsbruck (I, 577 m, region IA), Linz (L, 260 m, region NE), and Ljubljana (Lj, 299 m, region SE). The Alpine Space (gray area) and the national borders (bold black lines) are also indicated. Regions with missing fire data are crosshatched.

2.2. Weather type classification

Weather types are defined as atmospheric conditions over a large area, which remain relatively stable for at least a few days (Werner and Gerstengarbe, 2011). There are several methodologies to categorize weather conditions. At the German Meteorological Service, an objective weather type classification has been developed for climatological applications, which is described only briefly here. Further details are available in the original publication (Hess and Brezowski, 1952) as well as in Bissolli and Dittmann (2001). This classification is currently based on indices derived from numerical model runs with the global circulation model GME (40 km resolution). Meteorological criteria for this classification are the advection of air masses, the cyclonality and the humidity of the troposphere. Per definition the weather types have to last at least three days. The standard classification area is Germany and adjacent regions, but for our purpose we assumed that the types are also valid for the whole Alpine area. The categorization is carried out once a day and the results are published in the internet (www.dwd.de/GWL) and in miscellaneous reports of the Potsdam Institute for Climate Impact Research (Werner and Gerstengarbe, 2011). The current time series of the weather type data goes back to 1881 and comprises 29 types. These types can be further aggregated into broader groups depending on the flow direction (zonal, meridional, mixed) or on the cyclonality (cyclonic, anticyclonic).

The classification that we used (Table 1) is a modification of the original method. It is mainly based on the flow direction (first column, after Philipp et al., 2010) but since our purpose is a forest fire danger analysis, we additionally subdivided the types into cyclonic and anticyclonic groups (second column).

Hence, the final categorization comprised eleven types, including additionally a large-scale high pressure system over Central Europe (H), a class for a low pressure over Central Europe (L) and a less frequent (only 1% of all days) transitional type (T) mostly occurring when the circulation shifts from one situation to another.

Beside the cited H, L, and T, the types are labeled by a two-letter abbreviation (Table 1, column 3), the first letter indicating the flow direction (N for north, E for east, etc.) and the last letter the cyclonality (C – cyclonic, A – anticyclonic). The attribution of the obtained weather patterns to the eleven types defined by Hess and Brezowski (1952) is reported in the last column.

2.3. Forest fire danger indices

Forest fire danger rating systems are based on meteorological data and are used by land management and fire-fighting agencies to issue public warnings and to alert fire fighters and rescue agencies (Cheney and Gould, 1997). The variety of such fire danger indices is large and nearly every country affected by forest fires has developed its own system of fire risk assessment (e.g. Russia – Nesterov index (Nesterov, 1949), USA – National Fire Danger Rating System (NFDRS, Bradshaw et al., 1983), Australia – McArthur index (Noble et al., 1980), Sweden – Angstrom index (Chandler et al., 1983)). In this paper we used selected subindices of the renowned Canadian Forest Fire Danger Rating System (CFFDRS), first developed in the 1980s by Van Wagner (1987) and subsequently modified and adapted for use in many countries worldwide (Mexico, Italy, Portugal, Spain, New Zealand, etc.). In fact, although designed for local pine stands, the Canadian system has been demonstrated to work well in other parts of the world (Viegas et al., 1999; Camia et al., 2006). The whole Canadian system consists of three subindices for describing the fuel moisture conditions (FFMC – Fine Fuel Moisture Code, DMC – Duff Moisture Code, DC – Drought Code) and three sub-indices for fire behavior (ISI – Initial Spread Index, BUI – Build Up Index, FWI – Fire Weather Index). In the present study we focussed on the three moisture codes because of their suitability in describing the probability of forest fire ignition. The FFMC describes the moisture content of litter and other dried fuels and is an indicator of the relative ease of ignition of a layer of dry weight of about 0.25 kg m^{-2}. The DMC is a numerical rating of the

Table 1
The objective weather types of Hess and Brezowski (1952) categorized into eleven different types depending on flow direction and cyclonalilty. CE is the abbreviation for Central Europe.

Flow direction	Cyclonality	Abbreviation	Original definition	
Northeasterly, Easterly	Anticyclonic	EA	NEA	Northeasterly anticyclonic
			HFA	High Fennoscandia, CE anticyclonic
			HNFA	High Fennoscandia-Iceland, CE anticyclonic
	Cyclonic	EC	NEZ	Northeasterly cyclonic
			HFZ	High Fennoscandia, CE cyclonic
			HNFZ	High Fennoscandia-Iceland, CE cyclonic
Southerly	Anticyclonic	SA	SWA	Southwesterly anticyclonic
			SA	Southerly anticyclonic
			SEA	Southeasterly anticyclonic
	Cyclonic	SC	SWZ	Southwesterly cyclonic
			SZ	Southerly cyclonic
			TB	Low British Isles
			TRW	Trough Western Europe
			SEZ	Southeasterly cyclonic
Westerly	Anticyclonic	WA	WA	Westerly anticyclonic
	Cyclonic	WC	WZ	Westerly cyclonic
			WW	Maritime westerly
Northwesterly, Northerly	Anticyclonic	NA	NWA	Northwesterly anticyclonic
			NA	Northerly anticyclonic
			HNA	High Iceland, CE anticyclonic
			HB	High British Isles
	Cyclonic	NC	NWZ	Northwesterly cyclonic
			NZ	Northerly cyclonic
			HNZ	High Iceland, CE cyclonic
			TRM	Trough Central Europe
	High	H	HM	High Central Europe
	Low	L	TM	Low Central Europe, Cut off
			BM	Ridge Central Europe
		T	U	Transient

average moisture content of a layer of about 7 cm depth or 5 kg m^{-2}. It is also an indicator of the likelihood of fuel being ignited by lightning. The DC is the slowest reacting code and is an indicator for the average moisture content of a deep, organic layer weighing around 25 kg m^{-2}. The indices are calculated daily at noon (LST – local standard time) based on the meteorological input parameters air temperature [°C], relative humidity [%], 10 m wind speed [m/s], 24 h accumulated precipitation [mm] and on the previous days' output (for a detailed description of the respective calculations, see Van Wagner (1987)).

2.4. Alpine forest fire database

A major problem that arises when trying to record forest fires at the Alpine scale is the heterogeneity among countries in recording the data. First, competent authorities vary according to the country, second the definition of wildfires (e.g. thresholds for the burnt area, etc.) and forest area differ, finally the resulting archives are often not complete and information is sometimes patchy. In addition, differences in regional national languages pose an important barrier to creating a harmonized forest fire database. As a result an overall characterization and investigation of wildfires in the Alpine region has been lacking for a long time. To address this need, the European Union Alpine Space project ALPFFIRS (ALpine Forest FIRe warning System) was launched in 2009 with the aim of setting up a shared fire danger warning system. In the framework of this project a harmonized Alpine forest fire database was set up (Valese et al., 2011). At the current time (May 2012) the Alpine fire-dataset comprises 24,082 fire records burning 145,518 ha from 2001 to 2010 (2005–2010 in Germany).

It holds information about the fire size (burnt area), location (on a regional basis), and start/ignition day. Additional information such as ignition source or exact location is available for some regions and selected cases. Only fires larger than 0.1 hectares were considered since the minimum burnt area to be reported was different between the countries. In Germany some administrative districts are completely missing and for some regions in Austria and Switzerland the database is still being completed. The short time series of maximum 10 years does not allow us to study changing fire regimes due to rising temperatures (Conedera et al., 2009; Valese et al., 2011).

2.5. Statistical analysis

The statistical analysis is subdivided into three parts related to weather types, forest fire danger indices and observed fire data. Firstly, to describe the climatological occurrence of the eleven different weather types defined in Section 2.1, the relative frequency of the single weather types was calculated by dividing the number of days in that class by the total number of days in the 60 years under investigation (1951–2010). Days between two weather types were marked as transient and were classified separately. Additionally, this analysis was done for a winter (November–April) and a respective summer season (May–October) corresponding to a vegetative and non-vegetative season in forest fire analyses (Conedera et al., 2009; Valese et al., 2011). For consistency reasons, this definition was applied to the whole Alpine area despite large seasonal differences between the regions (e.g. in April some inner Alpine valleys may still be snow-covered, while the vegetation at southern exposed slopes in Italy or France is already fully developed). Seasonal differences in annual percentages of weather types were tested by Student's *t*-tests.

The meteorological forest fire danger (Section 2.3) was assessed by the three moisture indices of the Canadian Forest Fire Danger Rating System (FFMC, DMC, DC), which were calculated for the five representative stations on a daily basis. The box-plots of the index distributions (median, 75th and 95th percentile marks) were done separately for the eleven weather types. Since the three

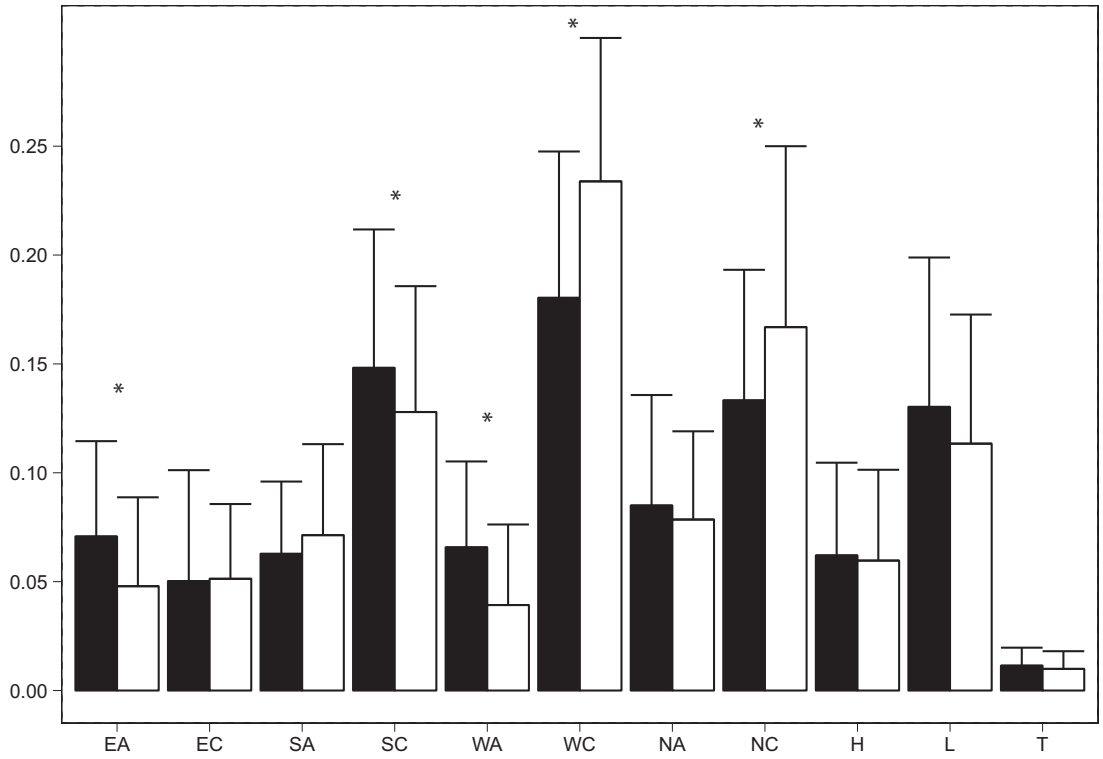


Fig. 2. Annual relative frequency of the eleven weather types from Table 1 (1951–2010). The black bars refer to the summer season (May–October) and the white bars to the winter season (November–April). Standard deviations of the frequency distributions (yearly basis) are displayed with error bars, significant differences ($p < 0.05$) between summer and winter are denoted with asterisks.

indices represent forest upper soil layers at different depths, a comparison between the distributions of the indices reveals different drying/wetting response times in the respective weather types. To test for differences in the index distributions between stations one-way analyses of variance (ANOVA, yearly basis) were conducted for each weather type, with Tukey's HSD for multiple comparisons where significant.

The observed forest fires in the Alps (2001–2010) aggregated on a national basis (since the exact location of the fire was not available for all cases) were compared to weather types with a relative frequency analysis for burnt area and number of fires. In order to enable a direct comparison among countries data on forest fire frequency were referred to units of forest area in the respective country and additionally, the resulting number was divided by the relative frequency of the weather types. For example, the weather type southerly anticyclonic (relative frequency of 0.06 in summer) was associated with 225 summer fires in the considered regions of Italy (284.34 km^2 forest area) over the 10-year period, which amounts to a relative number of 1.32 $km^{-2}\,yr^{-1}$. 939 ha area burnt resulted in a respective relative area burnt in this class of 5.50 $ha\,km^{-2}\,yr^{-1}$. The statistical analysis and the plots were performed in R (R Development Core Team, 2012).

3. Results and discussion

3.1. Weather types

The seasonal occurrence of the eleven weather types under investigation is displayed in Fig. 2. The most frequent weather type in both the summer and winter season was westerly cyclonic (WC) with a relative frequency of 21% over the whole year. This is not surprising since the Alps are situated exactly in the prevailing westerlies which range between 40°N and 60°N in Europe, depending on the season. In total, westerly weather types occurred more frequently than every fourth day (26%) in the time period 1951–2010. The next most frequent flow direction was north with a relative occurrence of 8% for the anticyclonic part and 15% for the cyclonic weather type. Less frequent were southerly and easterly winds with frequencies between 5% and 6%, 7% and 14%, respectively. For southerly wind directions cyclonic conditions were more frequent while easterly winds were more anticyclonic. In Central Europe low pressure systems (L) with a relative frequency of 11% occurred more often than high pressure systems (H) with 8%. Generally, averaged over all wind directions, the Alps are a region where cyclonic situations are significantly more frequent (67%) than anticyclonic situations (32%, $p < 0.001$). Cyclonic weather types are usually accompanied by rising air with the formation of clouds and likely precipitation while anticyclonic situations are normally characterized by fair weather. The surplus of cyclonic weather types can again be explained by the location in the prevailing westerlies which are characterized by very variable weather conditions and a high storm activity. The type T (transient), which contains all days which could not be assigned to a certain weather type, was rare (1%).

It is apparent from the frequency distributions for summer (May–October, black bars) and winter (November–April, white bars), that for most weather types the seasonal differences were not very strong. In particular, the classes EC, SA, NA, H and T were equally distributed between the two seasons (p-values between 0.4 and 0.9). The highest seasonal differences were found in the classes EA, SC, WA, WC and NC. For these weather types Student's t-test produced p-values below 0.05, for WA and WC even below 0.01.

In summer, almost every 5th day could be classified as westerly cyclonic, followed by the southerly weather type SC at 15% and the northerly situation NC at 13%. Prevailing low pressure systems (L) were also quite common in the Central European summer. In contrast, EC and SA were very rare weather types in summer at about

5% relative frequency and long lasting high pressure systems only occurred rarely (6%). In general, cyclonic weather was significantly more frequent in summer (64%) than anticyclonic (35%, $p < 0.05$).

In winter (i.e. the non vegetative season) the situation was very similar to summer. The weather type westerly cyclonic (WC) was also the most frequent (23%), followed by the northerly cyclonic (NC, 17%) type. In general, the proportion of cyclonic weather types between November and April (69%) was significantly higher than in summer ($p < 0.01$). It can be attributed to the more southerly frontal zone of the polar jet stream in winter. Since global pressure systems follow the sun, the zone with the lowest pressure in the Northern Hemisphere is situated between 50°N and 60°N in winter, while in summer this zone can be found further north (60–70°N). Hence, storm activity and also cyclonality in Central Europe are usually higher in winter than during the summer months. Interestingly, this pattern could not be found in the type of prevailing low pressure systems (L), which were more frequent in summer. This can be attributed to the pressure difference between northern and southern latitudes which is usually significantly higher in winter. This results in many small and fast moving low pressure systems across Europe in winter (mostly in conjunction with westerly winds). Due to strong insolation and low pressure differences in summer large prevailing low pressure systems often develop over the heated land mass in Central Europe.

3.2. Weather types and forest fire danger

3.2.1. FFMC

The distributions of forest fire danger indices as a function of the weather types are plotted in Fig. 3 for the five stations under investigation. Starting with the FFMC (first column), the fastest reacting moisture code in the Canadian system, remarkable differences between summer (gray bars) and winter (white bars) exist. At all five stations and for all weather types the index values were considerably higher during summer than winter (77.6 compared to 66.1 on average). This fact is due to the strong temperature dependency of the FFMC and most other forest fire danger indices. High temperatures and strong radiation accelerate the drying of the upper litter layer (of which the FFMC is representative) and hence the meteorological forest fire danger rises much faster after a precipitation event in summer than in winter. Furthermore, winter in Alpine valleys and plains are usually characterized by cold air and high relative humidity, which also reduces the calculated forest fire danger. Dew and rime effects are not considered in the Canadian system. However, this does not exclude that also in winter the index value can reach high levels in connection with long drought periods.

The highest index values occurred for the anticyclonic weather types EA, SA, NA and prevailing high pressure systems (H) which are usually associated with dry conditions and a lot of sunshine. On the other hand the cyclonic weather types EC, WC and NC had the lowest FFMC values due to a high relative humidity and frequent precipitation. Furthermore, summer days with the weather type L were also characterized by a relatively high forest fire danger, while in winter this class had only low FFMC values. A low pressure system in summer is usually characterized by a high convective activity with occasional sunshine that dries the upper litter layer very quickly after a rain shower. On the other hand in winter low pressure systems frequently cause wet and foggy conditions with no drying. One-way analyses of variance (ANOVA) based on a yearly basis revealed significant FFMC differences between the five stations ($p < 0.001$ for all weather types and both seasons). Also the Tukey's HSD for multiple comparisons revealed significant differences for all station combinations within a weather type and season ($p < 0.001$). The differences between stations were strongly dependent on flow direction. For example, weather type NC reveals very low median index values between 45 and 65 at the stations north of the Alps (Zurich, Linz) and also in the inner Alpine area (Innsbruck), while at the two stations south of the Alps (Locarno, Ljubljana) the FFMC ranges between 60 and 85, depending on the season. This underlines the ability of the Alps to act as a meteorological divide. While north of the Alps rising air caused precipitation and low temperatures, south of the Alps katabatic winds created dry conditions with a relatively high meteorological forest fire danger. The opposite occurred with the weather type SC where the station Locarno had considerably lower FFMC values than Linz or Innsbruck. Interestingly at Zurich low index values also appeared in connection with southerly cyclonic weather situations. This can be explained by its location in the western part of the Alps, because a southerly cyclonic flow is usually connected with a low pressure system over Western Europe which causes wet conditions in the Western Alps.

The differences between Northern and Southern Alps were considerably lower for the anticyclonic weather types because no precipitation usually occurs under these weather conditions. In general, the station with the highest FFMC values was Innsbruck where the median was about 78, while Zurich had a median FFMC of 66, i.e. about 15% lower. These differences can be explained by the topography. Zurich is situated north of the Alps where much rain falls (1100 mm) throughout the year (128 precipitation days) while Innsbruck is located in a Central Alpine valley. Such an inner Alpine area is sheltered from all sides and receives less precipitation (880 mm, 110 precipitation days). Additionally, in Innsbruck dry katabatic foehn winds occur regularly which result in an extremely high forest fire danger. At the stations south of the Alps relatively high FFMC values appeared throughout the whole summer while winter was rather rainy in these regions.

3.2.2. DMC

Descriptive statistics of the DMC values (Fig. 3, second column) reveal that the median in summer generally lay between 10 and 30 and the extreme values (95th percentile, black line) reached values between 40 and 80, strongly depending on station and weather type. The differences between summer and winter season were greater than for the FFMC which can be attributed to the deeper soil layer that is considered in the DMC. Hence, a few dry days in a row were sufficient to dry the upper layer of the forest litter which caused an increase of the FFMC, but the deeper soil was not dried sufficiently to change the DMC. Naturally the low temperatures were also responsible for the very low DMC values between November and April. As for the FFMC, Zurich had the lowest values and the station Innsbruck the highest. In summer very high DMC values up to 40 were recorded at the southern station Ljubljana due to long drought periods and high temperatures. At Innsbruck summer DMC exceeded 60 in connection with anticyclonic weather conditions. Comparing different weather types, the highest DMC values were connected with anticyclonic weather types (EA, WA, NA, H), while the cyclonic situations EC, WC and NC produced the lowest DMC values. However, the differences between the single weather types within the stations were not as large as for the fast-reacting FFMC. For instance, the median DMC values for many cyclonic weather types (EC, WC) which were characterized by strong regional differences in the FFMC analysis did not differ considerably between stations. Only Zurich had generally lower DMC values than the other stations. The reason for these smaller differences can again be found in the thicker layer considered in the DMC system. Since weather types do not usually last more than a few days in Central Europe, there is not enough time to dry the deeper forest soil layer. Similarly to FFMC, seasonal DMC significantly differed between stations for all weather types ($p < 0.001$).

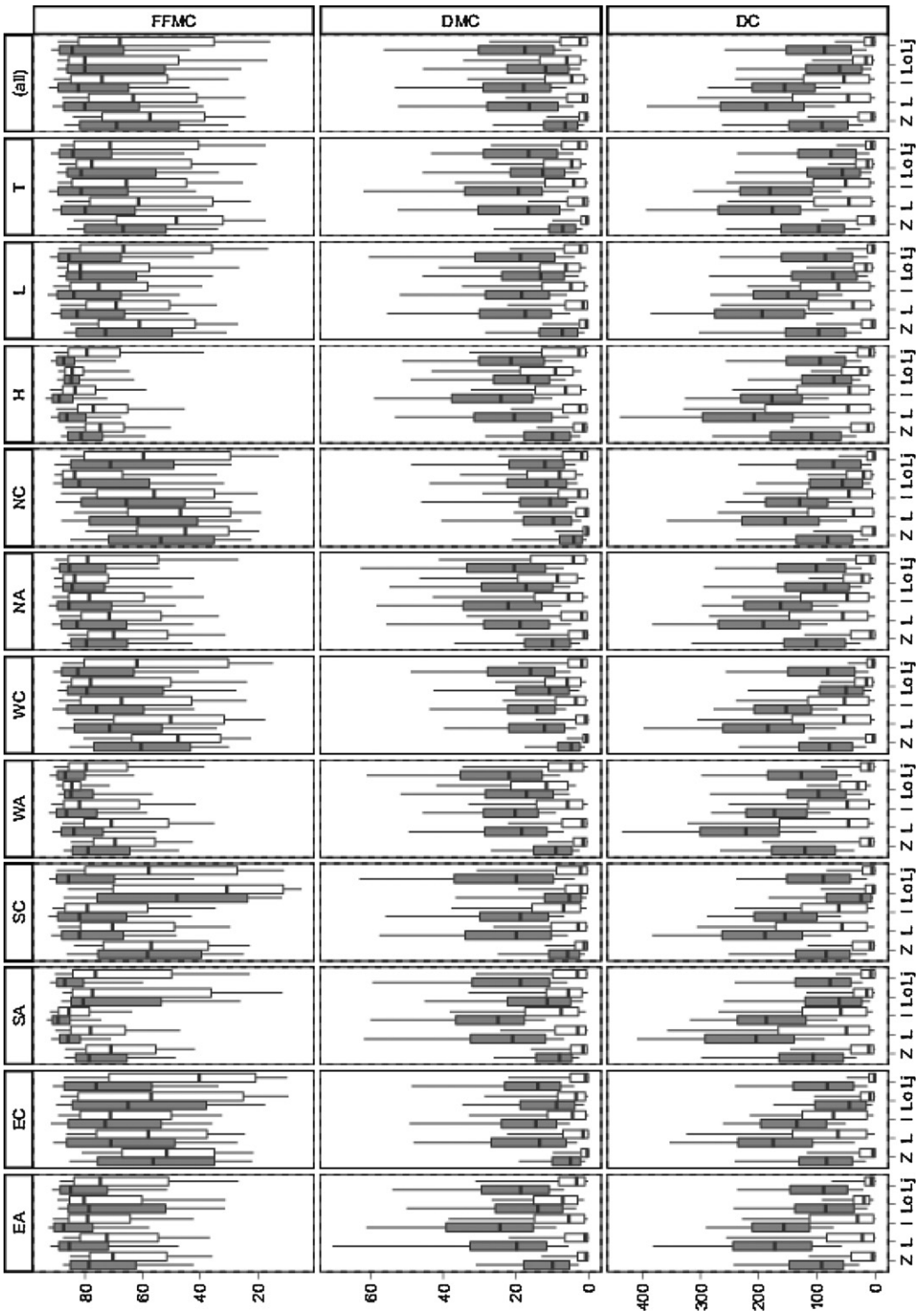


Fig. 3. Box-plots of the three forest fire danger indices FFMC, DMC and DC as a function of the weather types (abbreviations see Table 1). The five stations on the *x*-axis (Zurich, Linz, Innsbruck, Locarno, Ljubljana) are representative of the five climate regions in the Alps. Gray bars refer to the summer season, white bars to the winter season. The box is defined by 25th and 75th percentile with median shown as a bar, whiskers extend to the 5th and 95th percentile.

3.2.3. DC

The conclusions from the FFMC and DMC values are confirmed by the DC distributions in Fig. 3 (right column). This drought code of the Canadian system refers to a very deep layer of forest soil and hence drying processes are significantly slower than for the other two indices leading to considerable differences between summer and winter. The 75th winter percentile at most stations was even lower than the 25th summer percentile. The deep forest soil layer remained wet during winter due to low temperatures and weak evaporation. The ANOVA analyses confirmed these strong differences between stations for both seasons ($p < 0.001$). The driest conditions were recorded in summer at the stations Linz and Innsbruck with DC values of more than 400 for anticyclonic weather types. At the other Southern Alpine stations Locarno and Ljubljana considerably lower values appeared mostly likely linked to higher precipitation in summer due to a higher convective activity. The

Table 2
Number of forest fires and respective burnt area [ha] in the six Alpine countries, separately for winter and summer. The numbers in column four and five are totals in the period 2001–2010 (Germany 2005–2010), column six and seven give respective relative numbers per km^2 forest area and year.

Country	Abbreviation	Season	Number of fires	Burnt area [ha]	Rel. number of fires [km^{-2} yr^{-1}]	Rel. burnt area [ha km^{-2} yr^{-1}]
Switzerland	C	S	306	593.85	2.39	4.64
		W	294	1388.13	2.30	10.85
Germany	G	S	14	1.94	0.31	0.15
		W	19	9.98	0.40	1.09
Austria	A	S	662	262.46	1.68	0.67
		W	594	416.88	1.51	1.06
France	F	S	7118	57,806.54	16.11	130.87
		W	3126	8141.65	7.08	18.43
Italy	I	S	4478	20,908.03	15.75	73.53
		W	6564	51,116.03	23.09	179.77
Slovenia	S	S	472	3436.32	3.68	26.77
		W	427	1376.97	3.33	10.73

DMC values in Zurich were comparable to those in Ljubljana, which was not the case for the FFMC and the DC. Generally, a strong correlation between precipitation amount and the moisture content in a deep soil layer could be found. Hence, the rather dry stations Linz and Innsbruck which have mean annual precipitation below 900 mm, were characterized by significantly higher DC values than Zurich, Locarno or Ljubljana where precipitation averages more than 1100 mm. The differences between cyclonic and anticyclonic weather types were negligible. Weather conditions in Central Europe are simply too variable and change too fast to influence the moisture conditions in a deep forest layer. Only the extreme values (95th percentile) differed considerably between cyclonic and anticyclonic weather types. A closer look at these cases revealed that weather types that lasted 10 days or more were responsible for this feature.

3.3. Weather types and observed forest fires

3.3.1. Alpine forest fire regime

The fire statistics of the six countries participating in the European Union Alpine Space Program are shown in Table 2. Note that some regions in Italy (e.g. Liguria) and France (e.g. Cote d'Azur) are included in the EU – Alpine Space definition even though they show strong characteristics of a Mediterranean fire regime.

In the ten years between 2001 and 2010 a total of 145,518 ha of forest burnt in the study area. Italy and France were the two countries most prone to forest fires with more than 65,000 ha each accounting for more than 94% of the total area burnt. Beside the cited Mediterranean component, additional reasons for this fact are more favorable climatological conditions in the southern Alpine countries with high temperatures and long drought periods as mirrored also in the index values (see Fig. 3) and a relative large proportion of forest area. According to the forest database (http://epp.eurostat.ec.europa.eu) 48% of the total Alpine forest area can be found in France and Italy. A small confounding factor is the quality of the database since in Germany and Austria the consistent registration of forest fires is still in its infancy and some provinces like Freiburg or Tübingen in Germany and Liechtenstein are missing in this analysis.

The smallest area burnt was in Bavaria in southern Germany where only 44 fires occurred burning about 57 ha. The total Alpine area burnt in summer (57%) was higher than in winter (43%), but regional differences were large. While in Germany only 50% of the area burnt during the summer months, this fraction was considerably higher in the southern Alpine countries France (88%) and Slovenia (71%). In Switzerland and Austria the larger area burnt was during winter. This feature is known from literature (e.g. Zumbrunnen et al., 2011) and has been assigned to the rapid spreading character of the surface fires that occur in winter in the deciduous broadleaved forests. Seasonal differences in the number of forest fires were small (Table 2) with the bias of highest percentages of summer fires in France (69%) due to the important portion of the Mediterranean area in the depicted data, while in Italy only 41% of the fires occurred in summer. For all other countries the proportions in summer and winter were equal.

In total, the considered areas of Italy and France registered both more than 10,000 fires in the past decade, followed by Austria (1256) and Slovenia (899). In the Alpine part of Germany only 44 forest fires were observed which can be partly explained by the humid climate and the shorter observation period (2005–2010).

The average size of forest fires was very different in the countries. The largest average fire size occurred in France during summer (>8 ha), probably due to the Mediterranean characteristics of the area, while in Germany and Austria in both seasons the average fire size did not exceed 1.4 ha. North of the Alps average fire sizes were larger in summer whereas south of the Alps, especially in France and Slovenia, winter fires were much larger. Overall the average size of Alpine forest fires in the ten years under investigation was between 5.6 ha in winter and 6.4 ha in summer. Single large fires mainly occurred in connection with long drought periods and in inaccessible regions where fire fighting operations were difficult. The largest individual fire (11,580 ha) was registered on the 21st September 1990 in the Mediterranean part of France. This fire was extraordinary for Central Europe, but compared to extreme events in other fire prone countries like Australia or Canada where fires of more than 100,000 ha are not unusual, it was relatively small.

To reduce the effect of the different forest areas and the different observation lengths in the countries under investigation we also calculated relative numbers of fires and area burnt (Table 2 column six and seven). However, the results are substantially consistent with the total numbers. The highest number of forest fires per year and km^2 forest area could be found in France (on average 0.16 fires per ha in summer) and Italy (0.23 fires in winter), followed by Slovenia (0.04) and Switzerland (0.02). The ranking was the same for the relative burnt area in the last column of Table 2, with numbers between 0.15 ha of burnt area per km^2 forest area and year in the German summer and 180 ha km^{-2} yr^{-1} in the Italian winter.

3.3.2. Forest fires and weather types

Most forest fires have been observed in conjunction with anticyclonic conditions (EA, NA, H), especially in the two southern countries France and Italy, as reflected by the relative numbers in

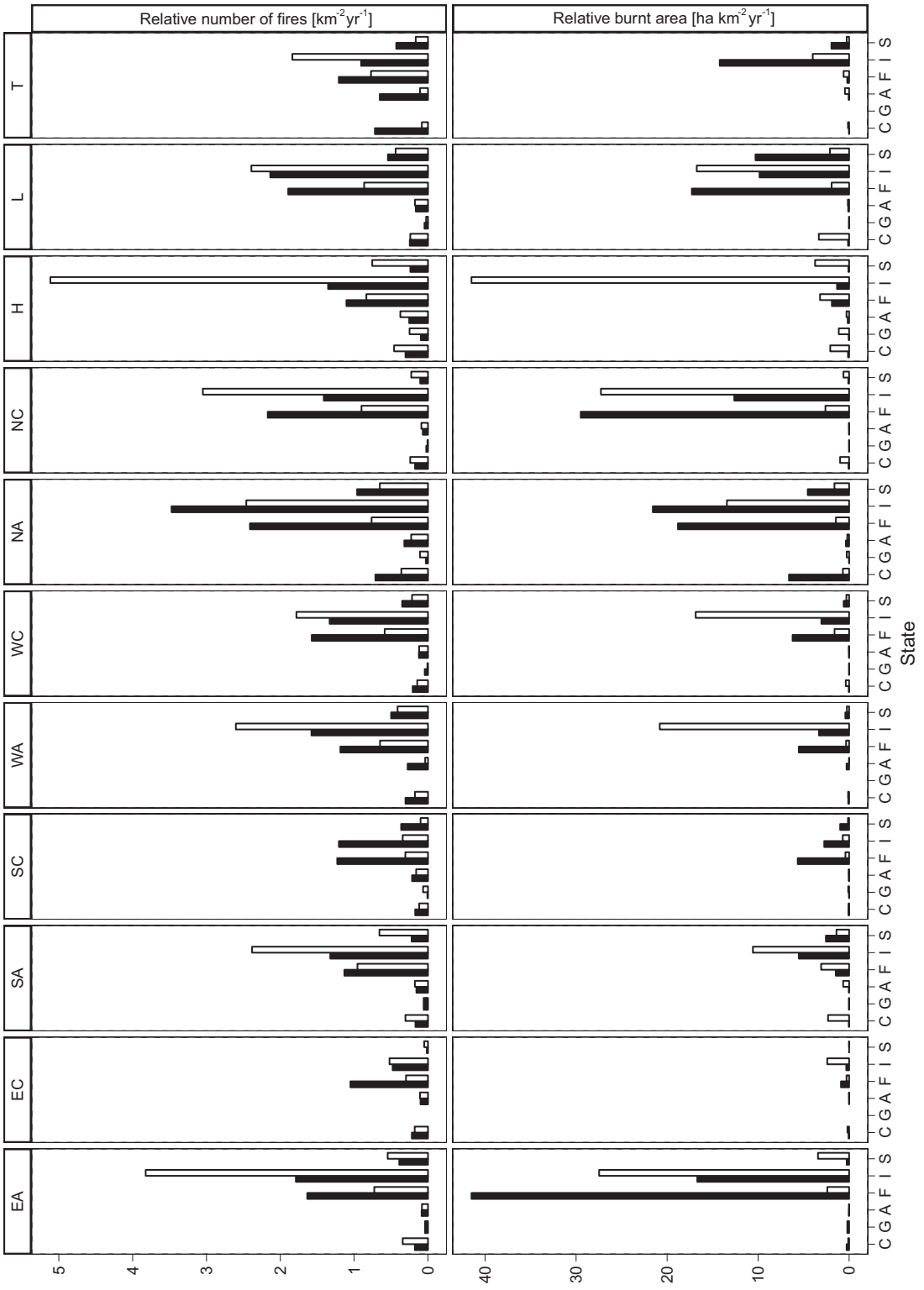


Fig. 4. Relative number of forest fires and respective area burnt in the six Alpine countries per weather type (i.e. divided by its relative occurrence). Black bars refer to summer, white bars to winter. Abbreviations are from Tables 1 and 2, the explanation of the relative numbers can be found in Section 2.5.

the first column of Fig. 4. This is not surprising since anticyclonic weather types are usually associated with dry conditions and a high meteorological forest fire danger (Fig. 3). However, on the other hand also in the cyclonic types NC and L a lot of forest fires occurred. A Student's *t*-test on a yearly basis revealed that the differences in the relative number of fires between cyclonic and anticyclonic weather types were significantly ($p = 0.02$) lower in summer than in winter. Cyclonic weather types are usually associated with humid conditions and low forest fire danger (see Fig. 3), but in summer cyclonic weather types are not always accompanied by large-scale precipitation. Since the convective activity in such weather conditions is usually very high, local precipitation can be exceptionally intense, while adjacent areas remain dry. With strong convection the frequency of lightning increases which can also act as

ignition source in the Alpine region. Furthermore, the wind speed in cyclonic weather types is higher, also increasing the fire danger and the fire rate of spread. A detailed analysis of the start date of the fire revealed that many fires occurred at the end of a dry anticyclonic period when the large-scale weather pattern has already turned to cyclonic. In such cases the wind speed was high and the desiccated forest litter was very prone to be ignited. This feature has already been identified in a German study in the 1960s (Baumgartner et al., 1967). The analysis might be partially biased by the fact that a high meteorological fire danger does not automatically imply a forest fire (Wastl et al., 2012), since the majority of fires in the Alps are ignited by human activities and that on days with extreme high meteorological fire danger public warnings and prohibitions (e.g. of open fires, barbecue, smoking, etc.) are given.

In Italy the highest relative number of forest fires occurred in persistent high pressure systems, the weather type EA and in connection with northerly winds (NA, NC). The latter can be attributed to special topographical characteristics in northern Italy. Since this region is sheltered by the Alps from the north (see Fig. 1) such weather types are frequently associated with strong katabatic winds in the lee of the Alps. These foehn winds are able to dry the soil very quickly and hence to increase the forest fire danger rapidly. In France the relative numbers are generally lower (higher forest area) with the weather types EA, NA, NC and L being most prone to forest fires. The patterns in Switzerland and Slovenia are quite similar to France and the contributions of the other two Alpine countries (Germany, Austria) in this diagram are very small. The transient class shows high relative numbers in nearly all countries.

The previous considerations are also valid for the burnt area during most weather types. The largest area burnt occurred on days with the anticyclonic weather type EA. Such flows often bring dry continental air masses to middle Europe which result in a high meteorological forest fire danger (Fig. 3). Also the northerly weather types (NA and NC) and persistent high pressure systems caused a high proportion of area burnt, especially in France and Italy. The absolute maximum in this figure occurred in Italy in connection with the weather type H, where a relative number of 43 ha km^{-2} yr^{-1} was reached in winter. In summer the relative numbers in this class were generally lower which can be attributed to convective precipitation in summer high pressure systems. In Austria and Switzerland the highest relative area burnt occurred with type NA, while in Slovenia the type L was top. As for the number of fires, also here the differences between cyclonic and anticyclonic weather types were significantly higher ($p = 0.04$) in winter than in summer.

Another interesting question that arose in this context was which weather type was associated with the biggest fires. Hence the area burnt was divided by the number of fires in each weather type. In Switzerland the largest average fires occurred in summer in connection with northerly anticyclonic flows (9.3 ha) and in winter with large-scale low pressure systems (13.9 ha). In Germany forest fires only appeared with southerly cyclonic weather types (foehn situations) and low pressure systems. In Austria large fires were usually associated with southerly flows and the maximum mean fire size in Slovenia occurred with low pressure systems (13.6 ha). The situation in the very fire prone France was characterized by strong seasonal differences in the fire size. In summer an easterly anticyclonic flow caused the biggest average fires at 25.4 ha, while in winter the largest average fires of about 3 ha occurred in conjunction with the anticyclonic weather types EA, SA and H. In Italy the analysis revealed clearly that foehn situations (WC and NC) and easterly anticyclonic flows caused the biggest fires in both seasons. The average size here was between 8 and 10 ha per fire. In general, averaged over all seasons and all countries, an easterly anticyclonic (EA) flow caused the largest forest fires in the Alps, which can be explained by the advection of very dry air masses from continental Europe. Northerly anticyclonic flows and persistent high pressure systems also lead to very large fires, even though the number of fires and the total area burnt were quite low in these types. This shows that the appearance of a forest fire does not depend solely on weather conditions, because human activities play a major role in the Alpine fire regime, but for a large forest fire to develop, the synoptic conditions have to be favorable.

4. Conclusions

In this study we investigate the association between atmospheric circulation types and forest fires in the Alpine area in Central Europe based on a combination of meteorological datasets (weather type classification, station data), calculated forest fire danger indices and observed forest fires.

The analysis of large-scale weather types revealed that westerly cyclonic flows were most frequent in Central Europe in the past 60 years, followed by northerly and southerly cyclonic types. The prevalence of cyclonic weather types could be attributed to large-scale rising mechanisms in the prevailing westerlies of the global circulation and is well known in literature (Bissolli and Dittmann, 2001).

The combination of weather types and calculated indices at five representative stations showed generally high correlations associated with flow direction and cyclonality. Elevated levels of forest fire danger indices predominantly occurred in conjunction with enduring anticyclonic situations or in the lee of the Alps due to katabatic foehn winds. The southern Alpine region was especially prone to forest fires – a fact that can mainly be related to favorable climatic conditions (Mediterranean climate influences) which have been described in many Alpine forest fire research papers (e.g. Zumbrunnen et al., 2009). Our conclusions were based on three different sub-indices of the Canadian Forest Fire Danger Rating System (FFMC, DMC, DC), the use of which has been tested for the complex ecosystem of the Alps by JRC (Joint Research Centre, http://ec.europa.eu/dgs/jrc/index.cfm) researchers (Camia et al., 2006). Since the three indices represent particular layer depths of forest soil, differences between the indices could be referred to drying mechanisms and response times in the soil. Differences between stations and weather types were smaller for deeper layers.

In order to base our results not only on meteorological evaluations, but also to link them to actually recorded fires, we decided to use observed forest fires in the Alps. However, the biggest challenge was the setup of a common forest fire database on the whole Alpine scale which was undertaken within the European Union project AlpFFIRS (Alpine Forest Fire Warning System, Valese et al., 2011). Due to different definitions (e.g. forest definition, threshold for forest fires, etc.) in the six Alpine countries and due to different lengths of the time series the datasets were reduced to a common period of ten years (2001–2010). Nevertheless, the dataset is patchy and has still to be completed because north of the Alps, where forest fires were not a big issue in the past, data from some regions and years are missing. Most forest fires (88%) occurred in the Southern Alpine countries of France and Italy, which have not only a more fire prone climate but also the best quality datasets of actual fires (Valese et al., 2011). At this point it has to be mentioned that some parts of Italy and France which are considered as Alpine in the EU definition show clear characteristics of a Mediterranean fire regime. Furthermore, human activities have also a very strong influence on Alpine forest fires (Zumbrunnen et al., 2011).

Most forest fires in the Alps occurred in connection with anticyclonic weather types (EA, NA, H) which confirms well with results of similar studies in other countries (Skinner et al., 2002; Hoinka et al., 2009; Rasilla et al., 2010). Surprisingly, many fires occurred also with cyclonic weather types when the calculated fire

danger was quite low. A further analysis of this fact revealed that many fires broke out at the end of a drought period before precipitation has started but the large-scale flow had already turned to cyclonic. This feature has already been identified by German researchers in the 1960s (Baumgartner et al., 1967). However, the biggest forest fires occurred in connection with large-scale high pressure systems and anticyclonic flows, which shows that for the development of large fires the meteorological conditions have to be favorable. In conclusion, for many regions a clear dependence of the occurrence of forest fires on certain weather types did not exist, which can partly be explained by the incomplete and patchy fire database in the Alps. Of course, also the strong human influence into the Alpine fire regime plays a decisive role in this context.

Further investigations on this topic should deal with shifts in the climatological frequency of large-scale weather types and associated changes in the occurrence of forest fires. Climate simulations by numerical models could be used to also generate outputs in terms of weather type distributions. However, investigations of temporal trends in fire statistics always have to be carried out with special care because human influences (e.g. leisure time activities, tourism, forestry management, etc.) have already changed a lot and will continue to change in the future.

Acknowledgements

The authors would like to thank the following institutions for providing the meteorological database: DWD (Deutscher Wetterdienst, Germany), ZAMG (Zentralanstalt für Meteorologie und Geodynamik, Austria), MeteoSwiss (Switzerland) and Environmental Agency of the Republic of Slovenia (Slovenia). The work also benefited from several national institutions and agencies and the European Union project MANFRED (http://www.manfredproject.eu) which helped to upset the extensive forest fire database. Financial support is acknowledged from the European Union through the Alpine Space ALPFFIRS project (no. 15-2-3-IT).

References

Auer, I., Bohm, R., Jurkovic, A., Lipa, W., Orlik, A., Potzmann, R., Schoner, W., Ungersbock, M., Matulla, C., Briffa, K., Jones, P., Efthymiadis, D., Brunetti, M., Nanni, T., Maugeri, M., Mercalli, L., Mestre, O., Moisselin, J.M., Begert, M., Muller-Westermeier, G., Kveton, V., Bochnicek, O., Stastny, P., Lapin, M., Szalai, S., Szentimrey, T., Cegnar, T., Dolinar, M., Gajic-Capka, M., Zaninovic, K., Majstorovic, Z., Nieplova, E., 2007. HISTALP – historical instrumental climatological surface time series of the Greater Alpine Region. Int. J. Climatol. 27, 17–46.

Baumgartner, A., Klemmer, L., Raschke, E., Waldmann, G., 1967. Waldbrände in Bayern 1950 bis 1959. Mitteilungen aus der Staatsforstverwaltung Bayerns 36.

Bissolli, P., Dittmann, E., 2001. The objective weather type classification of the German Weather Service and its possibilities of application to environmental and meteorological investigations. Meteorol. Z. 10, 253–260.

Bowman, D.M.J.S., Balch, J.K., Artaxo, P., Bond, W.J., Carlson, J.M., Cochrane, M.A., D'Antonio, C.M., DeFries, R.S., Doyle, J.C., Harrison, S.P., Johnston, F.H., Keeley, J.E., Krawchuk, M.A., Kull, C.A., Marston, J.B., Moritz, M.A., Prentice, I.C., Roos, C.I., Scott, A.C., Swetnam, T.W., van der Werf, G.R., Pyne, S.J., 2009. Fire in the Earth system. Science 324, 481–484.

Bradshaw, L.S., Deeming, J.E., Burgan, R.E., Cohen, J.D., 1983. 1978 NFDRS: technical documentation. USDA Forestry Service – Technical Report 39.

Camia, A., Barbosa, P., Amatulli, G., San-Miguel-Ayanz, J., 2006. Fire danger rating in the European Forest Fire Information System (EFFIS). In: Proc. Fifth International Conference on Forest Fire Research, Coimbra.

Chandler, C., Cheney, P., Thomas, P., Trabaud, L., Williams, D., 1983. Fire in Forestry – Forest Fire Behaviour and Effects. John Wiley & Sons, New York/Chichester/Brisbane/Toronto/Singapore.

Cheney, N.P., Gould, J.S., 1997. Fire growth and acceleration. Int. J. Wildland Fire 7, 1–5.

Conedera, M., Cesti, G., Pezzatti, G.B., Zumbrunnen, T., Spinedi, F., 2006. Lightning induced fires in the Alpine region: an increasing problem. In: Proc. Fifth International Conference on Forest Fire Research, Coimbra.

Conedera, M., Tinner, W., Neff, C., Meurer, M., Dickens, A.F., Krebs, P., 2009. Reconstructing past fire regimes: methods, applications, and relevance to fire management and conservation. Quaternary Sci. Rev. 28, 555–576.

Frelich, L.E., 2002. Cambridge Studies in Ecology. Forest Dynamics and Disturbance Regimes: Studies from Temperate Evergreen-Deciduous Forests, pp. 1–266.

Gatheron, J., Lavoine, J., 1950. Forest fires in the Landes of Gascony in 1949. Bull. Tech. Inform. Ingenieurs Serv. Agric., 123–134.

Giglio, L., van der Werf, G.R., Randerson, J.T., Collatz, G.J., Kasibhatla, P., 2006. Global estimation of burned area using MODIS active fire observations. Atmos. Chem. Phys. 6, 957–974.

Hess, P., Brezowski, H., 1952. Katalog der Grosswetterlagen Europas 33.

Hoinka, K.P., Carvalho, A., Miranda, A.I., 2009. Regional-scale weather patterns and wildland fires in central Portugal. Int. J. Wildland Fire 18, 36–49.

Krebs, P., Pezzatti, G.B., Mazzoleni, S., Talbot, L.M., Conedera, M., 2010. Fire regime: history and definition of a key concept in disturbance ecology. Theory Biosci. 129, 53–69.

Nesterov, V.G., 1949. Combustibility of the Forest and Methods for its Determination. USSR State Industry Press.

Noble, I.R., Bary, G.A.V., Gill, A.M., 1980. McArthur's fire-danger meters expressed as equations. Aust. J. Ecol. 5, 201–203.

Pezzatti, G.B., Zumbrunnen, T., Bürgli, M., Ambrosetti, P., Conedera, M. Fire regime shifts as a consequence of fire policy and socio-economic development: an analysis based on the change point approach. Forest Policy Econ., in press.

Philipp, A., Bartholy, J., Beck, C., Erpicum, M., Esteban, P., Fettweis, X., Huth, R., James, P., Jourdain, S., Kreienkamp, F., Krennert, T., Lykoudis, S., Michalides, S.C., Pianko-Kluczynska, K., Post, P., Alvarez, D.R., Schiemann, R., Spekat, A., Tymvios, F.S., 2010. Cost733cat – a database of weather and circulation type classifications. Phys. Chem. Earth 35, 360–373.

R Development Core Team, 2012. R: A Language and Environment for Statistical Computing. R foundation for Statistical Computing, Vienna.

Rasilla, D.F., Garcia-Codron, J.C., Carracedo, V., Diego, C., 2010. Circulation patterns, wildfire risk and wildfire occurrence at continental Spain. Phys. Chem. Earth 35, 553–560.

Reinhard, M., Rebetez, M., Schlaepfer, R., 2005. Recent climate change: rethinking drought in the context of forest fire research in Ticino, South of Switzerland. Theor. Appl. Climatol. 82, 17–25.

Skinner, W.R., Flannigan, M.D., Stocks, B.J., Martell, D.L., Wotton, B.M., Todd, J.B., Mason, J.A., Logan, K.A., Bosch, E.M., 2002. A 500 hPa synoptic wildland fire climatology for large Canadian forest fires, 1959–1996. Theor. Appl. Climatol. 71, 157–169.

Valese, E., Conedera, M., Vacik, H., Japelj, A., Beck, A., Cocca, G., Cvenkel, H., Di Narda, N., Ghiringhelli, A., Lemessi, A., Mangiavillano, A., Pelfini, F., Pelosini, R., Ryser, D., Wastl, C., 2011. Wildfires in the Alpine region: first results from the ALPFFIRS project. In: Proc. Fifth International Wildfire Conference, South Africa.

Van Wagner, C.E., 1987. Development and Structure of the Canadian Forest Fire Weather Index System. Canadian Forestry Service Forestry Technical Report 35.

Viegas, D.X., Bovio, G., Ferreira, A., Nosenzo, A., Sol, B., 1999. Comparative study of various methods of fire danger evaluation in Southern Europe. Int. J. Wildland Fire 9, 235–246.

Wastl, C., Schunk, C., Leuchner, M., Pezzatti, G.B., Menzel, A., 2012. Recent climate change: long-term trends in meteorological forest fire danger in the Alps. Agric. Forest Meteorol. 162–163, 1–13.

Werner, P., Gerstengarbe, F.W., 2011. Katalog der Grosswetterlagen Europas (1881–2010) nach Paul Hess und Helmut Brezowsky 119.

Zumbrunnen, T., Bugmann, H., Conedera, M., Burgi, M., 2009. Linking forest fire regimes and climate – a historical analysis in a dry inner Alpine valley. Ecosystems 12, 73–86.

Zumbrunnen, T., Pezzatti, G.B., Menendez, P., Bugmann, H., Burgi, M., Conedera, M., 2011. Weather and human impacts on forest fires: 100 years of fire history in two climatic regions of Switzerland. Forest Ecol. Manage. 261, 2188–2199.